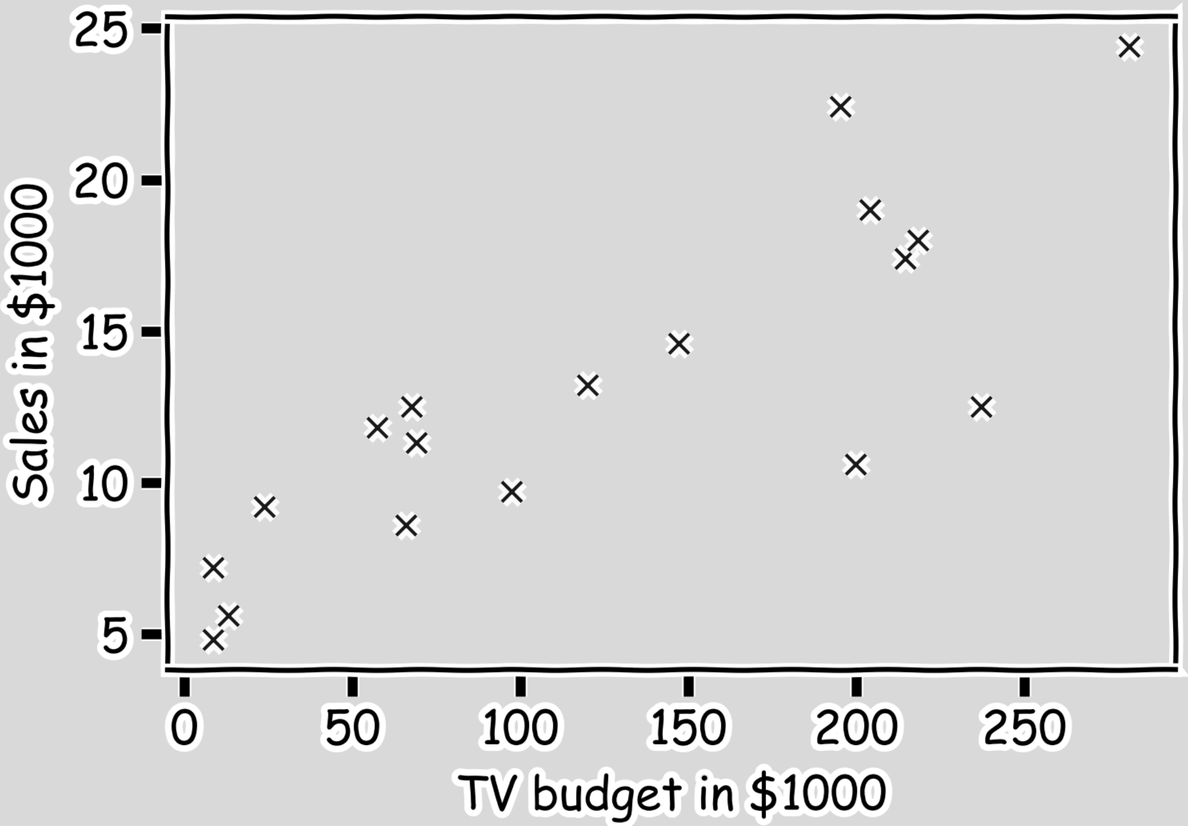
Error Evaluation & Model Fitness

Department of
ENGINEERING AND
SCIENCE EDUCATION
Clemson® University

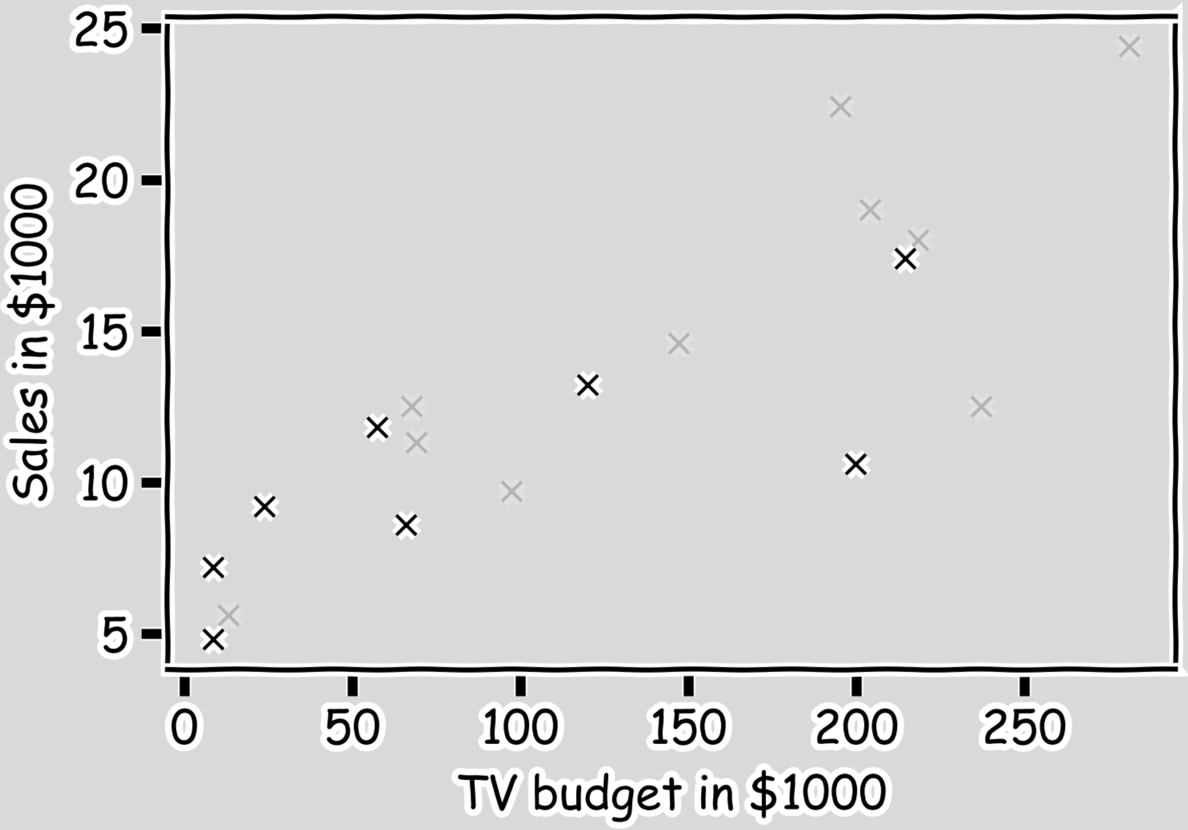
Error Evaluation

Start with some data.



Error Evaluation

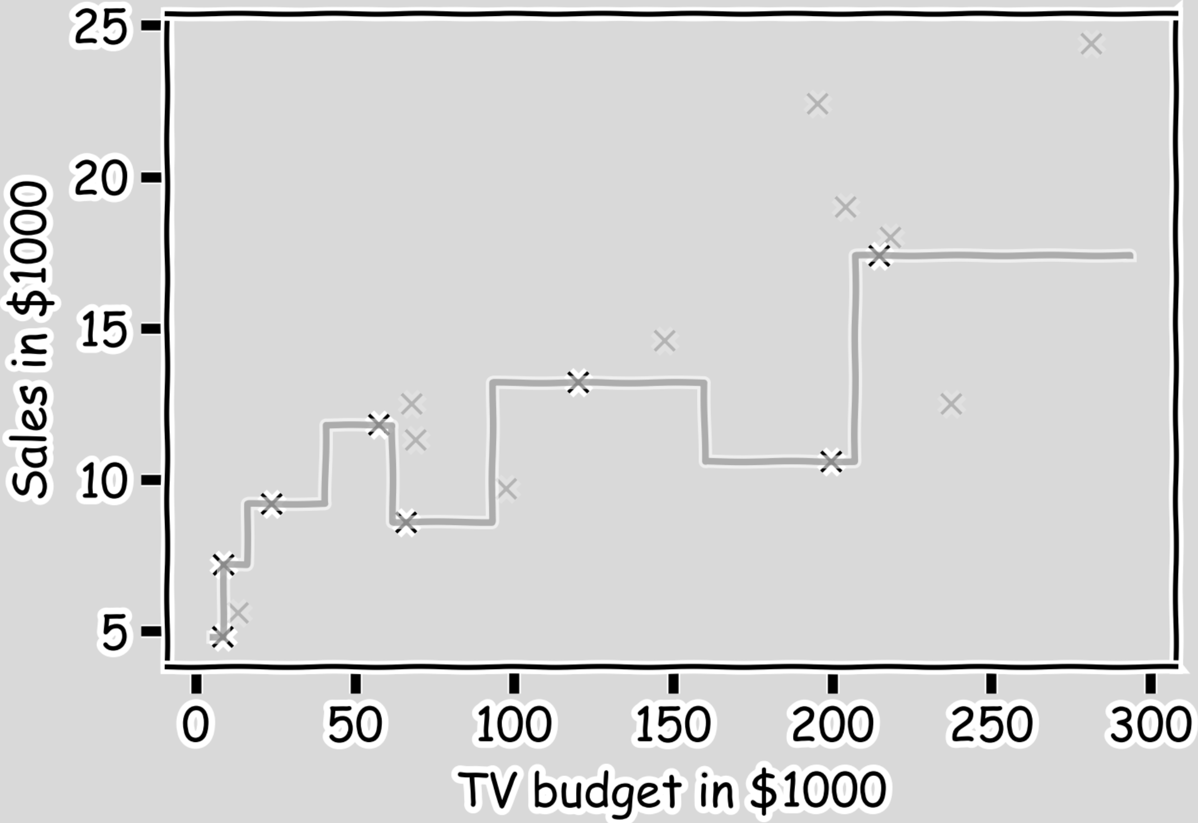
Hide some of the data from the model. This is called train-test split.



We use the train set to estimate $\hat{y}$, and the test set to evaluate the model.

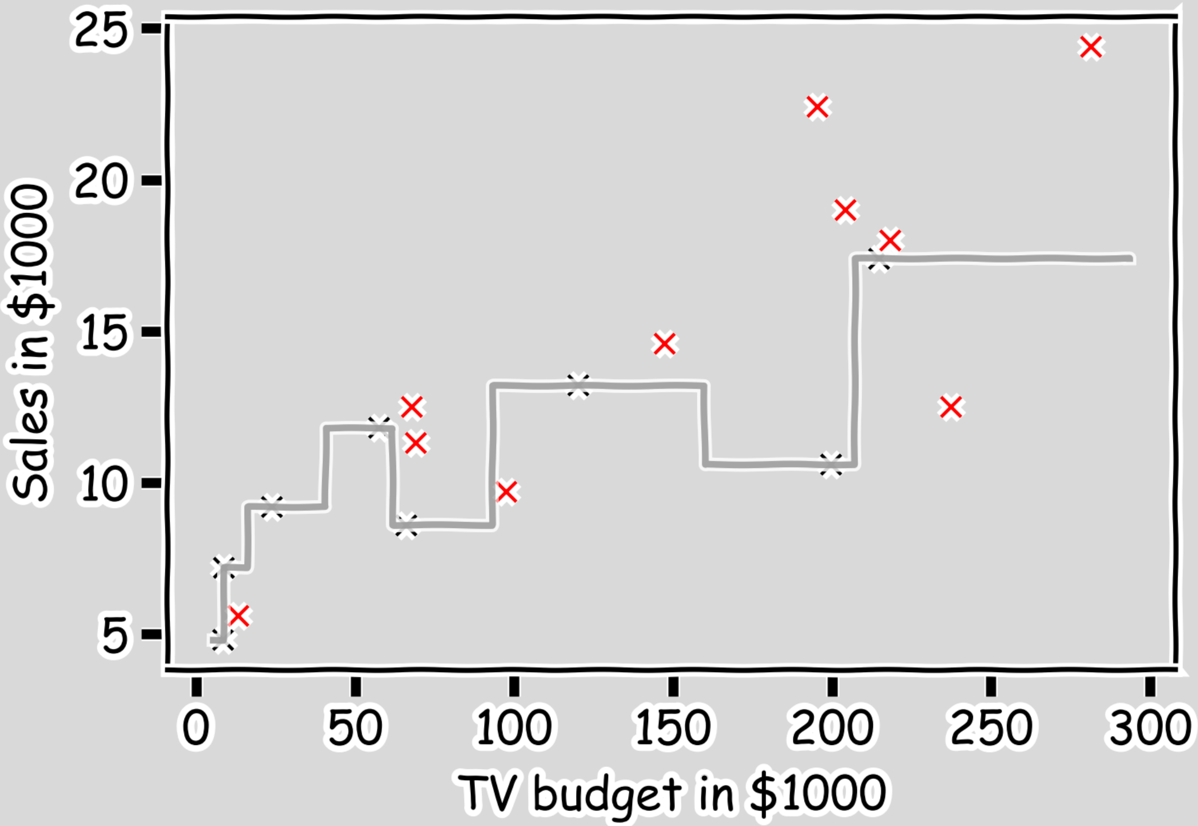
Error Evaluation

Estimate $\hat{y}$ for k=1 .



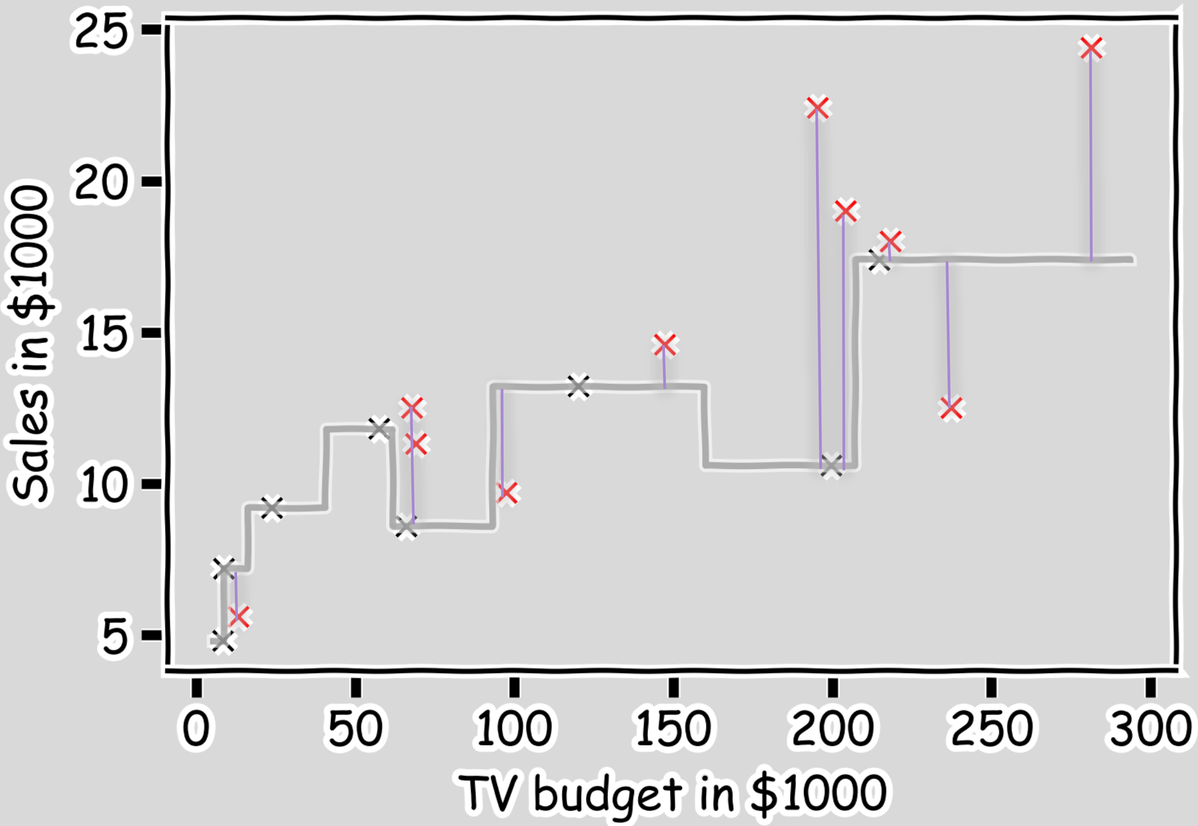
Error Evaluation

Now, we look at the data we have not used, the test data (red crosses).



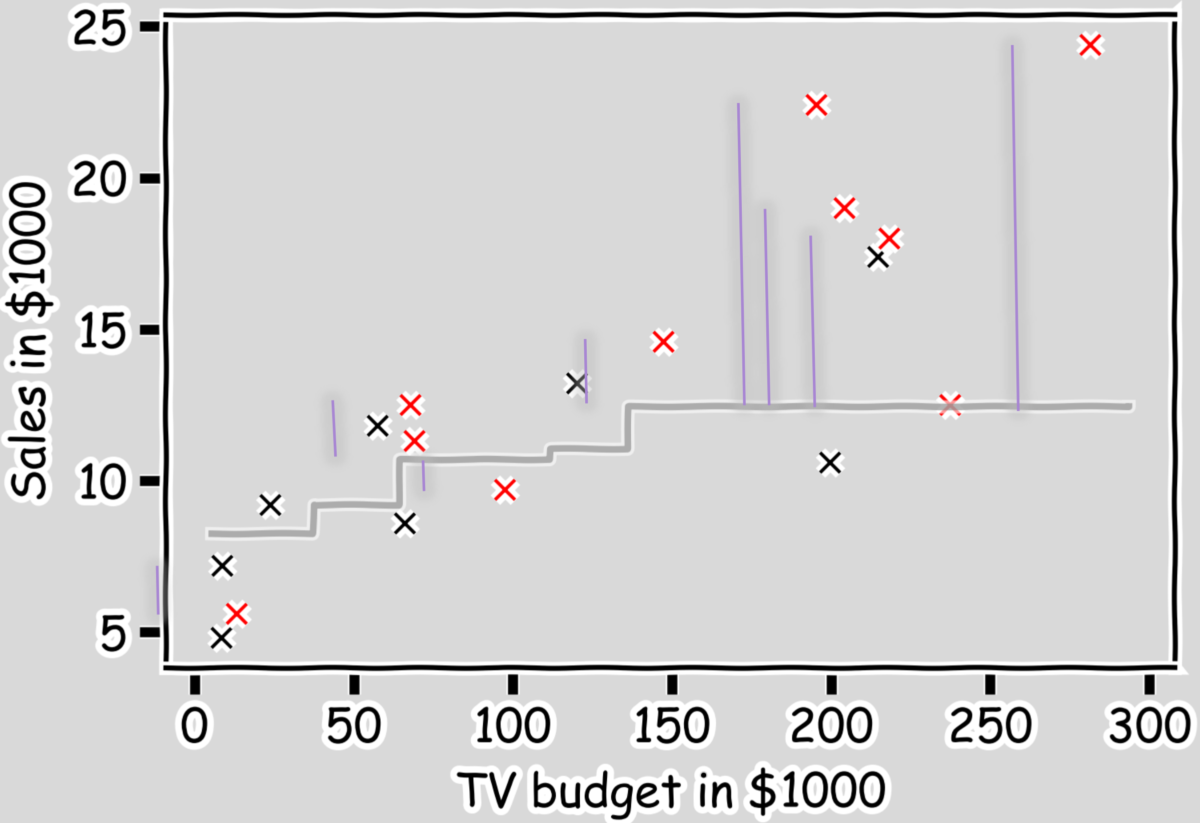
Error Evaluation

Calculate the **residuals** $(y_i - \hat{y}_i)$.



Error Evaluation

Do the same for *k*=3.



Error Evaluation

In order to quantify how well a model performs, we define a *loss* or *error function*.

A common loss function for quantitative outcomes is the **Mean Squared Error (MSE)**:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2$$

The quantity $y_i - \hat{y}_i$ is called a *residual* and measures the error at the *i*-th prediction.

Error Evaluation

Caution: The MSE is by no means the only valid (or the best) loss function!

Question: What would be an intuitive loss function for predicting categorical outcomes?

Note: The square **R**oot of the **M**ean of the **S**quared **E**rrors (RMSE) is also commonly used.

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})^2}$$

Things to Consider

Comparison of Two Models

How do we choose from two different models?

Model Fitness

How does the model perform predicting?

Evaluating Significance of Predictors

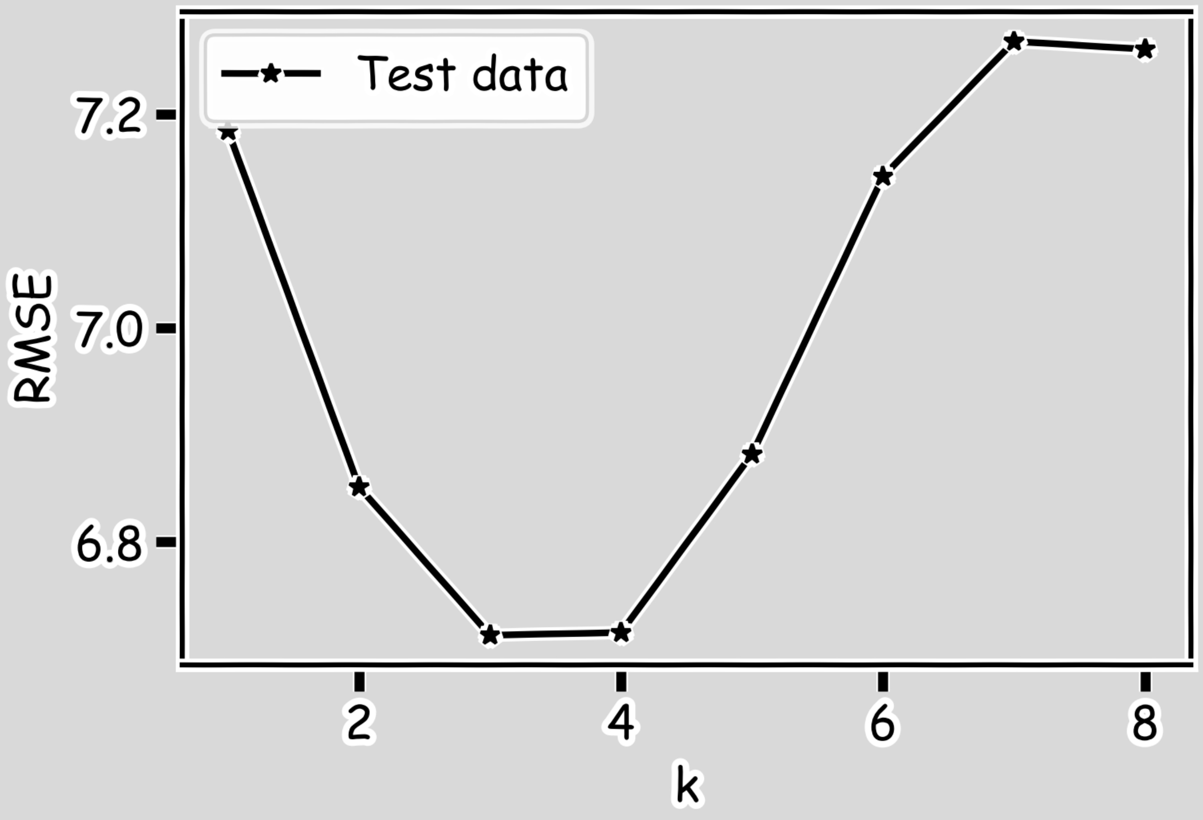
Does the outcome depend on the predictors?

How well do we know $\hat{f}$

The confidence intervals of our $\hat{f}$

Model Comparison

Do the same for all *k*’s and compare the RMSEs. *k*=3 seems to be the best model.



Things to Consider

Comparison of Two Models

How do we choose from two different models?

Model Fitness

How does the model perform predicting?

Evaluating Significance of Predictors

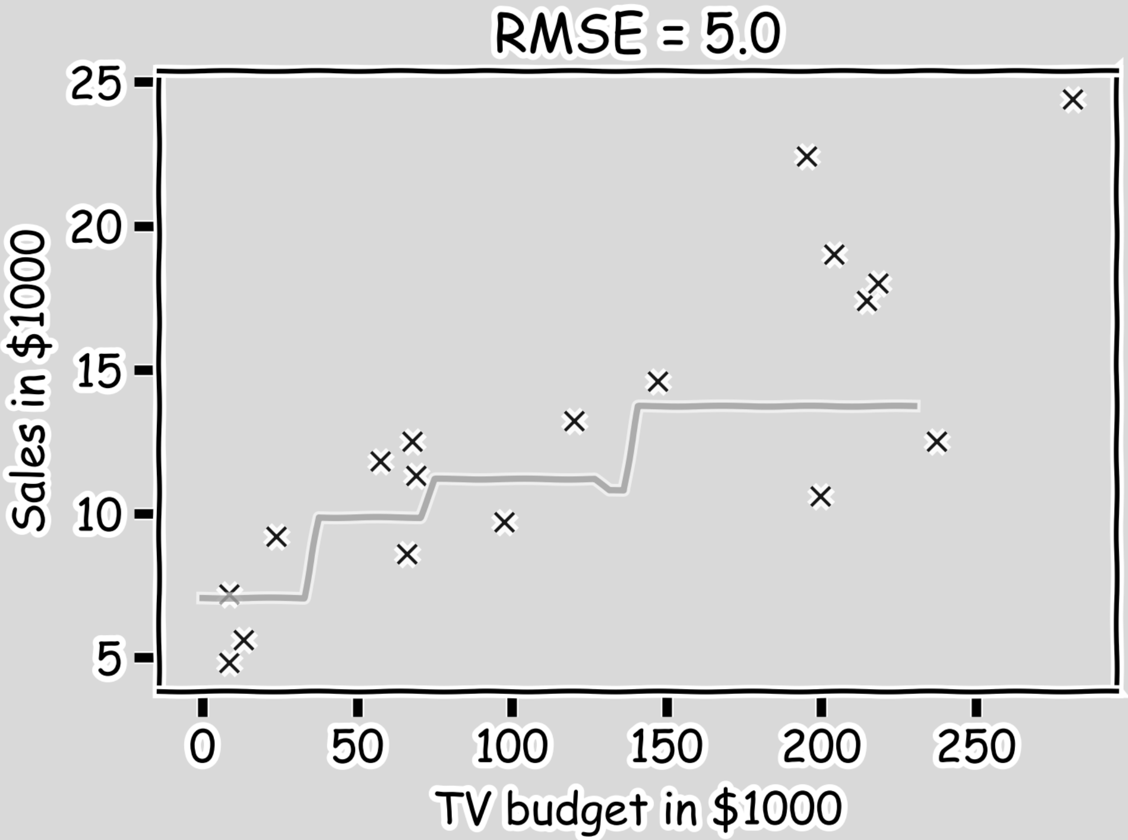
Does the outcome depend on the predictors?

How well do we know $\hat{f}$

The confidence intervals of our $\hat{f}$

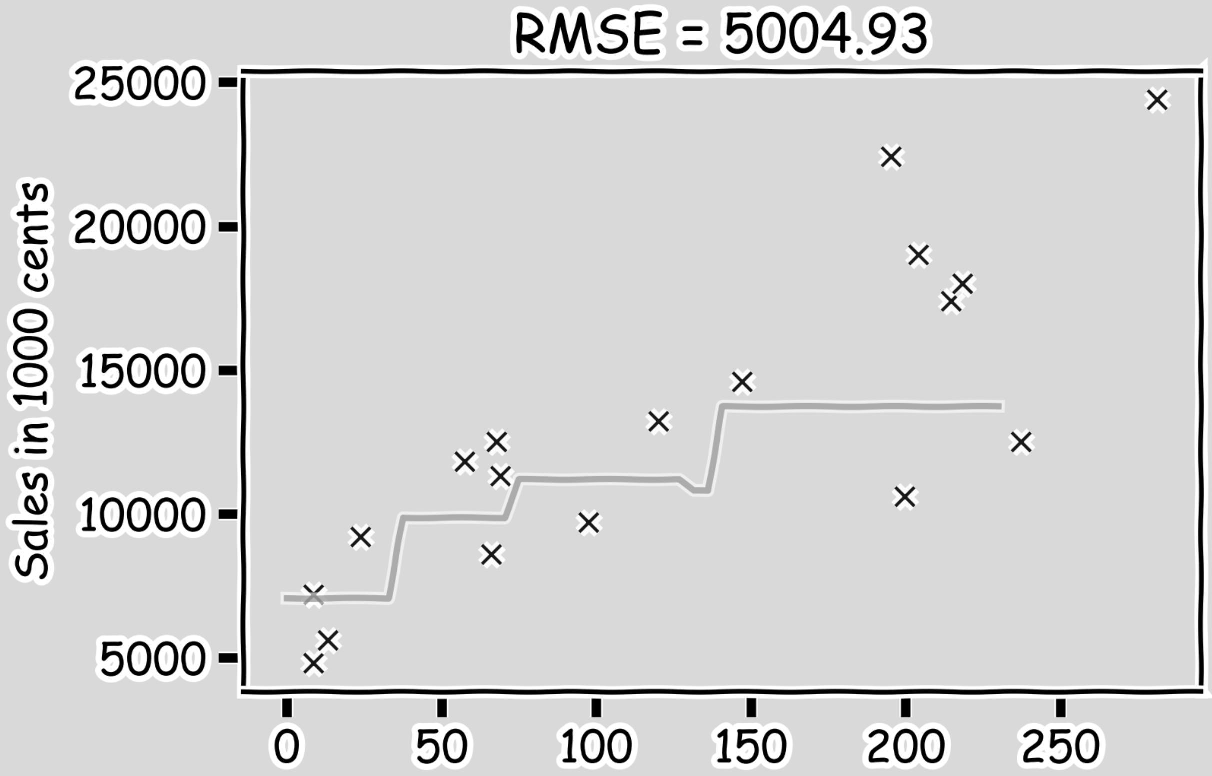
Model Fitness

For a subset of the data, calculate the RMSE for *k*=3. Is RMSE=5.0 good enough?



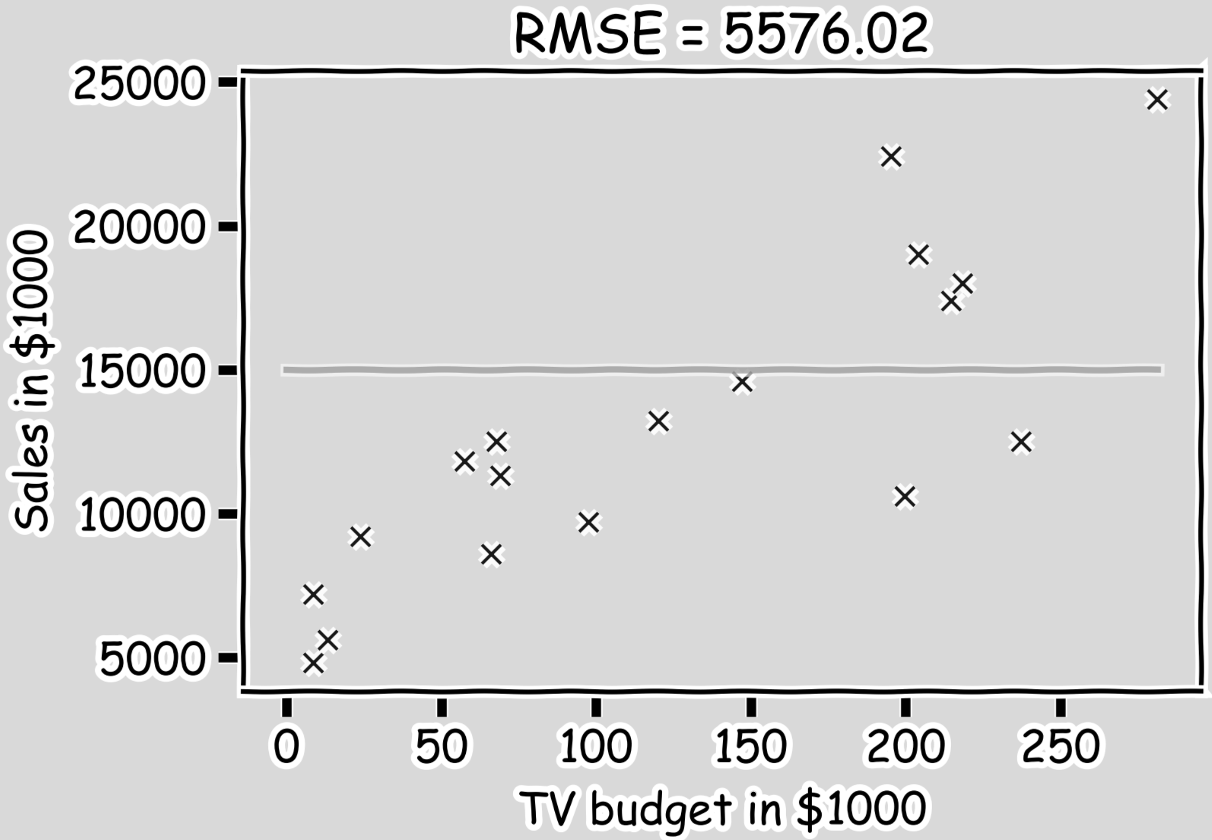
Model Fitness

What if we measure the Sales in cents instead of dollars?



Model Fitness

It is better if we compare it to something.



We will use the simplest model:

$$\hat{y} = \frac{1}{n}\sum_{i}^{n} y_i$$

R-Squared

- If our model is as good as the mean value, $\bar{y}$, then $R^2 = 0$
- If our model is perfect then $R^2 = 1$
- R^2 can be negative if the model is worst than the average. This can happen when we evaluate the model in the test set.

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (\bar{y} - y_i)^2}$$

Things to Consider

Comparison of Two Models

How do we choose from two different models?

Model Fitness

How does the model perform predicting?

Evaluating Significance of Predictors

Does the outcome depend on the predictors?

How well do we know $\hat{f}$

The confidence intervals of our $\hat{f}$

Things to Consider

Comparison of Two Models

How do we choose from two different models?

Model Fitness

How does the model perform predicting?

Evaluating Significance of Predictors

Does the outcome depend on the predictors?

How well do we know $\hat{f}$

The confidence intervals of our $\hat{f}$

Next lecture

Questions?